

IMO 2026 — Problem 4

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

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Approaches tried

- **Integral-multiple descent (successful).** When $180^\circ = N\theta$, regard all angles in units of θ . First force both children to have an integral angle, then repeatedly split an integral angle

m into 1 and $m - 1$.

- **Integer-entry interval lemma (successful).** For nonintegral $a+b+c = N$, after relabeling there is an integer k strictly between b and $a + b$; this specifies a legal cut whose two new opposite angles are $N - k$ and k .
- **Quotient-group invariant (successful).** Modulo $\theta\mathbb{Z}$, if the class of 180° is nonzero and all three current angle classes are nonzero, every cut has at least one child with the same property. Shan-Yu retains that child forever.
- **Computational sanity check.** Random normalized triangles for $2 \leq N < 30$ verified the interval construction, positivity of all child angles, and the formulas for the two forced integral angles.

Current best

Mulan can guarantee victory exactly for

$$\theta = \frac{180^\circ}{N} \quad \text{with an integer } N \geq 2.$$

For these values she first forces an integral multiple of θ to survive regardless of Shan-Yu's choice, and then decreases that multiple until it is θ . For every other θ , Shan-Yu starts with an equilateral triangle and uses a nonzero-class invariant modulo $\theta\mathbb{Z}$ to avoid θ forever.

Full proof

We prove both directions.

1. Sufficiency

Suppose

$$180^\circ = N\theta$$

for an integer $N \geq 2$. Measure all angles in units of θ , so the angles of every triangle sum to N .

We first record an elementary fact.

Lemma 1. Let $a, b, c > 0$ satisfy $a + b + c = N$, where $N \geq 2$ is an integer. If none of a, b, c is an integer, then the variables can be relabeled so that, for some integer k ,

$$b < k < a + b.$$

Proof. If one of the three numbers, say a , exceeds 1, take either of the other two numbers as b . Since b is not an integer,

$$b < [b] < b + 1 < b + a.$$

Thus $k = [b]$ works.

It remains to consider the case in which all three numbers are at most 1. As none is an integer, all are in fact less than 1, so $N = a + b + c < 3$. Since $N \geq 2$ is an integer, $N = 2$. Choose any one of the numbers as c and call the other two a, b . Then $b < 1$ and

$$a + b = 2 - c > 1.$$

Hence $k = 1$ works. $\square$

Now consider any current triangle. If it already has an angle θ , Mulan has won. If one of its normalized angles is an integer m , then necessarily $1 \leq m \leq N - 1$. We will shortly explain how Mulan wins from such a position.

It therefore remains only to handle a triangle whose normalized angles a, b, c are all nonintegral. Relabel its vertices using Lemma 1, and choose an integer k such that

$$b < k < a + b.$$

Mulan cuts from the vertex having angle a to the opposite side, choosing the cut so that the part of angle a adjacent to the vertex having angle b is

$$x = k - b.$$

The inequalities above give $0 < x < a$, so the ray lies strictly inside the angle and meets the interior of the opposite side. Thus this is a legal cut.

In the child adjacent to the angle b , the angle at the point of the cut is

$$N - b - x = N - k,$$

whereas in the other child the angle at the point of the cut is

$$N - c - (a - x) = N - c - a + k - b = k,$$

because $a + b + c = N$. Therefore, no matter which child Shan-Yu retains, its triangle has an angle that is a positive integral multiple of θ . (Both k and $N - k$ are strictly between 0 and N , as they are angles of nondegenerate child triangles.)

Finally, suppose a current triangle has an angle $m\theta$, where m is an integer with $1 \leq m \leq N - 1$. If $m = 1$, Mulan wins. If $m \geq 2$, she cuts from that vertex along the interior ray which splits the angle $m\theta$ into angles

$$\theta \quad \text{and} \quad (m - 1)\theta.$$

Such a ray meets the interior of the opposite side, so the cut is legal. One child has an angle θ and the other has an angle $(m - 1)\theta$. If Shan-Yu keeps the first, Mulan wins at the next stopping check; if he keeps the second, the positive integer m has decreased by 1. Repeating this argument forces victory after at most $m - 1$ further cuts. This proves sufficiency.

2. Necessity

Suppose now that

$$180^\circ \notin \theta\mathbb{Z}.$$

We show that Shan-Yu can prevent victory indefinitely.

Work in the additive quotient group

$$G = \mathbb{R}/(\theta\mathbb{Z}),$$

and denote the class of a real angle r by $[r]$. In particular, let

$$t = [180^\circ] \neq 0.$$

Call a triangle *safe* if none of its three angles has class 0 in G , equivalently, if none is an integral multiple of θ .

Shan-Yu can initially choose an equilateral triangle. It is safe: if $[60^\circ] = 0$, then $60^\circ = n\theta$ for some integer n , which would give $180^\circ = 3n\theta$, contrary to the assumption.

We prove that after every cut of a safe triangle, at least one child is safe. Let the classes of the parent angles be a, b, c . Then

$$a, b, c \neq 0, \quad a + b + c = t \neq 0.$$

Suppose Mulan cuts from the vertex with angle class a , splitting that angle into classes u and v , where $u + v = a$. The two child triangles have angle-class triples

$$(u, b, c + v) \quad \text{and} \quad (v, c, b + u).$$

Indeed, for example, the third angle of the first child has class

$$t - b - u = (a + b + c) - b - u = c + v,$$

and similarly for the second child.

Assume for contradiction that neither child is safe. Since $b, c \neq 0$, the first child can contain a zero class only if

$$u = 0 \quad \text{or} \quad c + v = 0,$$

and the second can contain a zero class only if

$$v = 0 \quad \text{or} \quad b + u = 0.$$

There are four possible pairings. If $u = v = 0$, then $a = u + v = 0$. If $u = 0$ and $b + u = 0$, then $b = 0$. If $c + v = 0$ and $v = 0$, then $c = 0$. Each contradicts safety. In the sole remaining case,

$$c + v = 0 \quad \text{and} \quad b + u = 0,$$

so

$$t = a + b + c = (u + v) + b + c = (b + u) + (c + v) = 0,$$

again a contradiction. Hence at least one child is safe.

After every cut, Shan-Yu retains a safe child. Since an angle equal to θ has class 0, a safe triangle never triggers Mulan's victory. Thus Mulan cannot guarantee victory when $180^\circ \notin \theta\mathbb{Z}$.

Combining necessity and sufficiency, and using $0^\circ < \theta < 180^\circ$, the desired values are exactly

$$\theta = \frac{180^\circ}{N} \quad (N = 2, 3, 4, \dots).$$